Weather Data Analysis, Forecasting & Visualization

End-to-end weather analytics combining a **real-time API feed** (OpenWeatherMap) with **historical analysis and forecasting** on 96K hourly observations (2006–2016), served through an interactive Streamlit dashboard.

Results

Next-hour temperature forecasting on a temporal holdout (19,257 test rows):

Model	MAE	RMSE
Naive (persistence)	0.841	1.210
Ridge	0.547	0.685
Gradient Boosting	**0.116**	**0.182**

Gradient boosting reduces MAE by **86%** over the naive baseline.

What's inside

- **Part 1 — Real-time collection:** `WeatherCollector` class pulling current conditions for 7 cities via the OpenWeatherMap API
- **Part 2 — Historical data:** 96,453 hourly observations, cleaning and validation
- **Part 3 — EDA:** monthly temperature climatology, daily cycle analysis, feature correlations
- **Part 4 — Feature engineering:** calendar features, 1h/24h/7-day lag features, 24h rolling means and standard deviations
- **Part 5 — Forecasting:** naive / Ridge / gradient boosting with a strict temporal train/test split (no shuffling), MAE & RMSE evaluation, feature importances, forecast-vs-actual visualization
- **Part 6 — Dashboard:** Streamlit app with Live, Historical Trends, and Forecast tabs

Run it

Kaggle

1. Open the notebook in Kaggle or Colab
2. Add the Historical Hourly Weather Data dataset (or upload `weatherHistory.csv`)
3. Run all cells top to bottom

4. For the live feed, add your OpenWeatherMap API key (free tier) — never commit keys to the repo

Tech

Python · pandas · scikit-learn · Plotly · Streamlit · OpenWeatherMap API